

9.2 – Disadvantages

Dependence on Data Quality:

- The accuracy of insights depends entirely on the quality of the input data.
- Incorrect, incomplete, or outdated data can lead to misleading results.

Limited Predictive Capability:

- Tableau primarily focuses on visualization and descriptive analytics.
- Advanced predictive analysis requires additional tools or machine learning integration.

Performance Issues with Large Datasets:

- Very large EV datasets may cause slower dashboard loading and rendering times.
- Complex visualizations can impact system performance.

Requires Tableau Knowledge:

- Users need basic knowledge of Tableau to create, customize, and maintain dashboards effectively.
- Training may be required for new users.

Licensing Costs:

- Full Tableau features require paid licenses.
- This may increase project costs for organizations and researchers.

Limited Real-Time Data Processing:

- Real-time monitoring depends on continuous data connectivity and refresh configurations.
- Delays in data updates can affect decision-making.

Customization Constraints:

- Certain advanced visualization requirements may be difficult to implement using Tableau's built-in features alone.
- Custom scripting options are limited compared to full software development solutions.

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

Data Security Concerns:

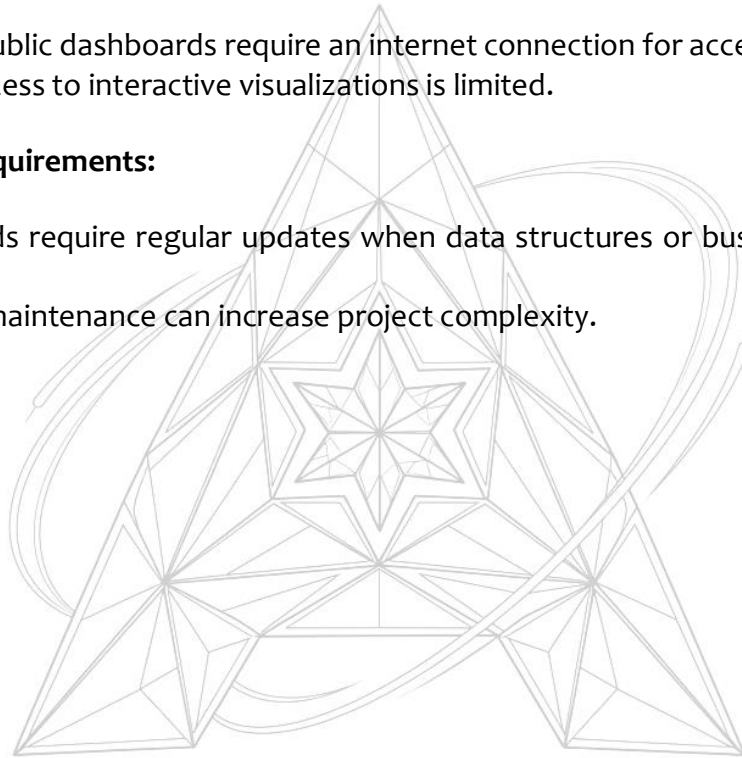
- Sharing dashboards online may expose sensitive EV operational data if proper security measures are not implemented.
- Access control and permissions must be carefully managed.

Internet Dependency:

- Tableau Public dashboards require an internet connection for access.
- Offline access to interactive visualizations is limited.

Maintenance Requirements:

- Dashboards require regular updates when data structures or business requirements change.
- Ongoing maintenance can increase project complexity.



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